MODELS	GPT-4	CHATGPT	INSTUCTGPT	COHERE	ALPACA	VICUNA	BAIZE	WIZARDLM
ORDER (COH.)	0.17_F	0.38_F	0.24_L	0.33_F	0.82_L	0.32_F	0.95_L	0.64_L
ORDER (DIV.)	0.14_F	0.45_F	0.22_L	0.23_L	0.76_L	0.52_F	0.83_L	0.68_L
EGOCENT. (COH.)	0.78	0.58	0.28	0.27	0.18	0.27	0.02	0.14
EGOCENT. (DIV.)	0.80	0.54	0.29	0.41	0.18	0.18	0.04	0.09
SALIENCE (COH.)	0.56	0.63	0.66	0.60	0.47	0.53	0.49	0.53
SALIENCE (DIV.)	0.57	0.69	0.70	0.65	0.49	0.59	0.50	0.52

Table 8: Comparison on the ORDER benchmark considering diverse evaluation perspectives. For visual clarity, we only display the bias ratio with the highest proportion and denote with subscript x_F or x_L for first- or last-ordered bias, respectively.

MODELS	GPT-4	CHATGPT	INSTUCTGPT	COHERE	ALPACA	VICUNA	BAIZE	WIZARDLM
ORDER	0.17_F	0.38_F	0.24_L	0.33_F	0.82_L	0.32_F	0.95_L	0.64_L
ORDER (TIE)	0.15_F	0.43_F	0.0	0.08_L	0.0	0.0	0.81_L	0.47_L
TIE (%)	0.01	0.0	0.88	0.33	0.95	0.99	0.0	0.04
EGOCENTRIC	0.78	0.58	0.28	0.27	0.18	0.27	0.02	0.14
EGOCENTRIC (TIE)	0.77	0.60	0.04	0.25	0.02	0.0	0.08	0.16
SALIENCE	0.56	0.63	0.66	0.60	0.47	0.53	0.49	0.53
SALIENCE (TIE)	0.55	0.67	0.06	0.35	0.01	0.0	0.50	0.48

Table 9: Comparison on the ORDER benchmark considering ties. For visual clarity, we only display the bias ratio with the highest proportion and denote with subscript x_F or x_L for first- or last-ordered bias, respectively.

B.5 Significance of Results

We adapt two statistical hypothesis tests based on the random bias threshold for the Order bias (first-order and last-order) benchmark in Table 10. Since we have binary outputs (bias, not biased), we conduct a two-sample Z test of proportions to determine the significance of each proportion of biased evaluations from each automatic evaluator with the random baseline. We conduct the test with the null hypothesis defined to be that "evaluator X is just as likely to make the mistake of flipping its preference according to the order of the response to the first-order as the random baseline" or equivalently:

H_0: the mean of Evaluator X for first-order bias is not any different from random selection.

On almost all of the ORDER benchmarks, the proportions of biased evaluations are statistically significant from ones by the random score. Notably, the p-values are critically small (z-scores are blown up) due to our large sample size. Also, we note that the p-value is actually not statistically significant for last-order bias in InstructGPT; however, the first-order proportions are statistically significant, indicating that one must consider the test for both positions to get the full picture of the evaluator's tendencies in reference to the random baseline. For example, if both first-order and last-order were not statistically significant from the random proportions, we might find that the evaluator is "unbiased," but the following may also undermine the capabilities of the automatic evaluator reduced to just random choice.

B.6 LLM Performance and Agreement

We detail the general agreement between machine preferences as similarly conducted in the human-machine correlation study. Figure 5 visualizes the average Rank-Based Overlap between LLMs. We find that LLMs in their own size group (excluding the smallest size group) have a relative agreement with each other. For example, models in the largest size group ($>100B$) are more in agreement amongst themselves than with models from other size groups. Furthermore, we also show the average valid response rate from different bias promptings in Table 12. We gather the proportion of valid responses by post-processing each "eval-gen" via pattern matching. After post-processing, we then label each output as a valid or invalid response, such that if a response is valid, we give one point to the preferred system.

B.7 Model Size

We conduct a supplementary experiment analyzing the impact of each bias for different models scaled by size in Table 13. We present results from a range of model sizes with LLAMAV2 and VI-

MODEL	FIRST ORDER Z-SCORE	FIRST ORDER P-VALUE	LAST ORDER Z-SCORE	LAST ORDER P-VALUE
GPT-4	8.45	$2.82e{-}17$	26.55	$2.65e{-}155$
CHATGPT	-15.61	$6.68e{-}55$	32.43	$9.50e{-}231$
INSTRUCTGPT	13.04	$7.08e{-}39$	1.17	$2.4e{-}1$
LLAMAV2	-33.12	$1.30e{-}240$	17.97	$3.53e{-}72$
LLAMA	-40.84	0	15.36	$2.95e{-}53$
COHERE	-10.02	$1.20e{-}23$	8.98	$2.59e{-}19$
FALCON	-57.36	0	25.61	$1.24e{-}144$
ALPACA	30.67	$1.29e{-}206$	-61.45	0
VICUNA	-12.49	$8.44e{-}36$	7.63	$2.29e{-}14$
OPENASSIST	-37.27	$4.84e{-}304$	13.93	$3.92e{-}44$
MISTRAL	-20.09	$9.13e{-}90$	32.56	$1.63e{-}232$
OLMO	-54.54	0	22.08	$4.47e{-}108$
BAIZE	35.46	$1.99e{-}275$	-71.53	0
KOALA	-21.60	$1.77e{-}103$	12.18	$4.15e{-}34$
WIZARDLM	20.65	$9.93e{-}95$	-40.48	0
MPT	-30.71	$4.81e{-}207$	16.49	$4.62e{-}61$

Table 10: Significance test scores for ORDER bias (first and last order preference) for each evaluator compared to the random baseline. We see almost every model shows significant results with $\alpha = 0.05$.

CUNA. Interestingly, we see that the valid response rate within LLAMAV2 goes down as the model size is scaled up, but the impact of each bias greatly increases as the model size is scaled down (with the exception of SALIENCE BIAS). On the implicit bias benchmarks, LLAMAV2 exhibits more robust performance with the proportion of responses affected by each bias SALIENCE BIAS in which longer responses are much more strongly preferred. For the induced bias benchmarks, a similar trend is viewed in which the effect of each bias on the model as an evaluator is dampened in correlation to the model scale. On the contrary, VICUNA exhibits a stronger valid response rate as the model size is scaled; however, certain implicit biases are much more amplified, such as ORDER BIAS and SALIENCE BIAS. For implicit biases, VICUNA tends to prefer itself when actual model names are used as size is scaled smaller while tending to prefer much more verbose responses as model size is scaled higher. Across the induced biases, VICUNA performs more resiliently proportionally to scale, although still strongly influenced by BANDWAGON EFFECT but much less affected by ATTENTIONAL BIAS. We include another visualization correlating the overall performance on each of the bias benchmarks with model size for the main results in Figure 2a.

B.8 N-Rankwise setting: $N = 4$

We show the results and average rankings between four different models representing each of the different size groups: CHATGPT ($>100B$), FALCON ($>40B$), ALPACA ($>10B$), VICUNA ($<10B$).

For the experimental setup, we conduct a smaller study, generating 100 responses from each of the 4 different LLMs using the Databricks Dolly15k dataset (Conover et al., 2023) via the same instruction prompt template from Appendix C and the same evaluation prompt template from the ORDER bias.

We only employ this setting under the order bias setting in order to validate the complexity of the task that modern (smaller) LLMs aren't capable of performing yet. We perform each experiment by randomizing the order of each list of responses and prompt each LM-as-evaluator to order the list from best to worst (top to bottom) according to the same criterion as the pairwise study (providing the instruction/sample reference). Furthermore, we also track ORDER bias, calculated by the proportion of responses in which the first (randomly) placed model was also ranked first by the evaluator.

As viewed in Table 14, we find that most models besides the closed-source API models (e.g. OpenAI) have trouble generating a proper rank list for even an $N = 4$ setting. This may be due to the increased complexity of the task (Dziri et al., 2023) where the ranking of N generations may become much more difficult as N gets larger (since the task complexity increases).

	GPT4	CHATGPT	INSTRUCTGPT
Ego	0.78	0.58	0.28
Longer Ego	0.64	0.75	0.43
Shorter Ego	0.36	0.25	0.56

(a) Model Performance Comparison (>175B)

	LLAMAV2	LLAMA	COHERE	FALCON
Ego	0.06	0.0	0.27	0.05
Longer Ego	0.18	0.0	0.68	0.6
Shorter Ego	0.81	0.0	0.32	0.4

(b) Model Performance Comparison (>40B)

	ALPACA	VICUNA	OPENASSIST
Ego	0.18	0.27	0.15
Longer Ego	0.38	0.4	0.71
Shorter Ego	0.62	0.59	0.29

(c) Model Performance Comparison (>10B)

	MISTRAL	OLMO	BAIZE	KOALA	WIZARDLM	MPT
Ego	0.3	0.0	0.02	0.48	0.14	0.21
Longer Ego	0.64	0.0	0.0	0.56	0.54	0.83
Shorter Ego	0.36	1.0	0.0	0.44	0.46	0.17

(d) Model Performance Comparison (<10B)

Table 11: Additional comparisons examining the proportion of EGOCENTRIC samples where the (self-preferred) model's generation was longer/shorter than the other generation.

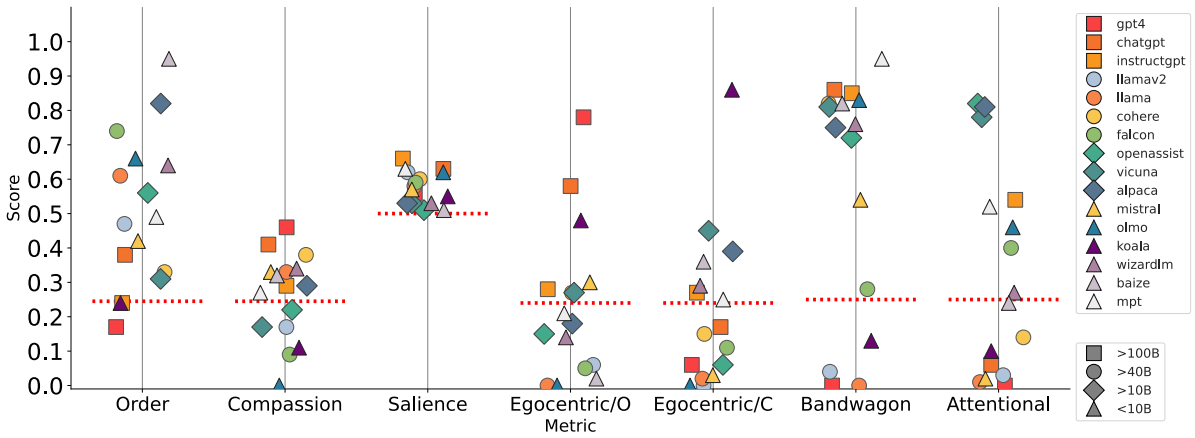

Figure 4: Proportion of responses that were labeled bias for each bias benchmark. We visualize the distribution of the 15 models tested that varies by the y-axis. The red dashed line indicates the RANDOM threshold for each bias benchmark that serves as a litmus between biased and unbiased LMs-as-evaluators. The spread on the x-axis is randomly distributed for visual clarity.